

Requirement Analysis

Functional Requirements:

Custom Table: The system must store all request data in a dedicated custom table

Student Form: Students must be able to submit a new book request through a form

Auto Request Number: Every request must automatically receive a unique, system-generated tracking number

Date Validation: The system must reject any request where the issue date is earlier than today

Status Tracking: Every request must have a status field with values: Pending, Approved, Rejected

Default Status: New requests must default to "Pending" status automatically

Email Notification: The librarian must receive an automatic email notification whenever a new request is submitted

Reporting: The system must provide visual reports showing request data (by status and as a list)

Dashboard: A single dashboard must combine reports into one view for quick monitoring

Non-Functional Requirements

Usability: The form must be simple, clean, and easy for students to fill without training

Performance: Business rules and notifications must execute in real time (under 1 second)

Security: Role-based access must separate student permissions from librarian/admin permissions

-Reliability: Validation logic must consistently prevent invalid data from being saved

Scalability: The table and logic should support a growing number of requests without redesign

Data Requirements — Fields Needed

- Student Name
- Student Email
- Book Title
- Author
- Issue Date
- Return Date
- Request Number **(auto-generated, read-only)**
- Status **(choice: Pending / Approved / Rejected)**
- Librarian Notes

Roles and Permissions

Admin Role (Librarian):

- Full access — Create, Read, Write, Delete

User Role (Student):

- Can Create, Read, and Write their own requests
- Cannot Delete requests (only librarian/admin can)

Tools and Technology Used

- ServiceNow Developer Instance (free tier)
- System Definition module — for Tables, Columns, and Business Rules
- Flow Designer (Workflow Studio) — for notification automation
- Reports module — for data visualization
- Performance Analytics / Dashboard module — for combined reporting view

Assumptions

- Each student submits requests individually (no bulk/group requests)
- Email notifications are sent only to a single librarian/admin in this version
- The developer instance is used for build and testing, not a production rollout

Constraints

- Developer instance has limited email-sending capability — verified via Flow Designer execution logs instead of live inbox delivery in some cases
- No integration with an external library catalog system in this version